

From Regions to Hierarchies: Fast, Context-Aware Off-Road Path Planning

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Abstract—Dense grids capture terrain detail but make kilometer-scale queries expensive; flat semantic abstractions speed up search but do not support localized contextual updates to map state. We address both limitations through two complementary representation layers. CLEAR converts co-registered landcover and elevation rasters into reusable convex semantic–geometric regions. HOPHY organizes this class of regions into a typed hierarchy whose contextual memberships identify affected terrain and accelerate repeated planning. Across 9–100 km² maps, CLEAR makes repeated queries $2.0\times$ – $19.6\times$ faster than raw-grid A* with 6.7% path-cost overhead. HOPHY obtains $68\times$ – $1280\times$ lower median query time with 0.00% median cost difference from pixel A* and sub-200 ms replacement planning under localized rain and blockage. A physical Jackal experiment evaluates the real-world feasibility of executing the planned mission. Together, these results show that explicit semantic–geometric regions can support scalable global path planning, localized contextual updates, and feasible real-world mission execution.

I. MOTIVATION

Semantic scene graphs and open-set maps support reasoning beyond geometry [1], [2], [3], while recent work extends semantic mapping to outdoor scenes and uncertain off-road perception [4], [5]. For a ground robot, however, terrain meaning is inseparable from continuous geometry and operating context. A landcover class, slope, weather condition, and vehicle type jointly determine whether a region is traversable. Mission autonomy also queries the same map repeatedly during task allocation and must revise routes when context changes.

Dense grids retain terrain detail but make kilometer-scale repeated search expensive. Flat abstractions accelerate search but do not encode which map state a contextual event affects. Incremental planners efficiently repair a search after changed edges are supplied [6], [7], but identifying those edges remains a semantic grounding problem. We therefore ask:

Research question: How can an outdoor semantic map preserve mobility-relevant terrain structure while supporting repeated global path-planning queries and localized adaptation under contextual change?

We answer through a two-layer representational argument. CLEAR addresses what the reusable semantic–geometric map units should be. HOPHY addresses how that class of units should be organized and associated with contextual events so that mission-level queries remain fast and map repair remains localized. Figure 1 summarizes this progression from terrain inputs to persistent context-aware planning state. We contribute:

- a shared region interface connecting semantic map construction to maintained planning state,
- complementary layers for reusable terrain units, hierarchical queries, and localized contextual updates, and
- mission-scale evidence spanning digital maps, repeated planning, contextual replanning, and outdoor robot execution.

II. A SHARED SEMANTIC–GEOMETRIC INTERFACE

Both layers operate on regions that can be expressed through the interface

$$R_k = (\mathcal{P}_k, \pi_k, \ell_k, \mathcal{N}_k), \quad (1)$$

where $\mathcal{P}_k$ is a connected polygon, π_k is a local elevation model, ℓ_k is a terrain-semantic label, and $\mathcal{N}_k$ is its adjacency list. This interface separates persistent map structure from a cost model. Distance, grade, terrain class, learned traversability, and agent-specific mobility may therefore change without rebuilding the spatial decomposition.

A. CLEAR: constructing reusable map units

CLEAR (Connected Landcover Elevation Abstract Representation) receives co-registered elevation $Z(p)$ and landcover $L(p)$. Boundary-Seed Decomposition allocates a fixed seed budget between semantic transitions, ranked by local label entropy, and elevation-flat interiors. Voronoi tessellation produces connected convex cells. Recursive plane fitting then subdivides cells whose elevation error exceeds a tolerance, producing the regions in Eq. (1) and their adjacency graph.

Across Wharton State Forest (9 km²), Humphreys Peak (50 km²), and Mount Rainier (100 km²), CLEAR answers repeated queries $2.0\times$ – $19.6\times$ faster than raw-grid A*. Its sparse decomposition introduces 6.7% path-cost overhead. In BeamNG.tech, CLEAR produces 5–8.6% shorter executed paths than Quadtree with 100% completion across the evaluated tasks. These results establish a reusable and interpretable map layer, while also exposing the fidelity cost of a flat sparse abstraction.

B. HOPHY: organizing units under context

HOPHY (Hierarchical Off-Road Planning using Hypergraphs) organizes the same *class* of semantic–geometric regions into three levels. Fine regions retain local terrain attributes. Geometric–semantic nodes merge compatible fine regions while preserving boundary connectivity. Coarse regions group these nodes for long-distance search.

Typed hyperedges associate sets of geometric–semantic nodes with terrain, agent, or mission context. Rain and blockage hyperedges identify affected terrain, agent hyperedges

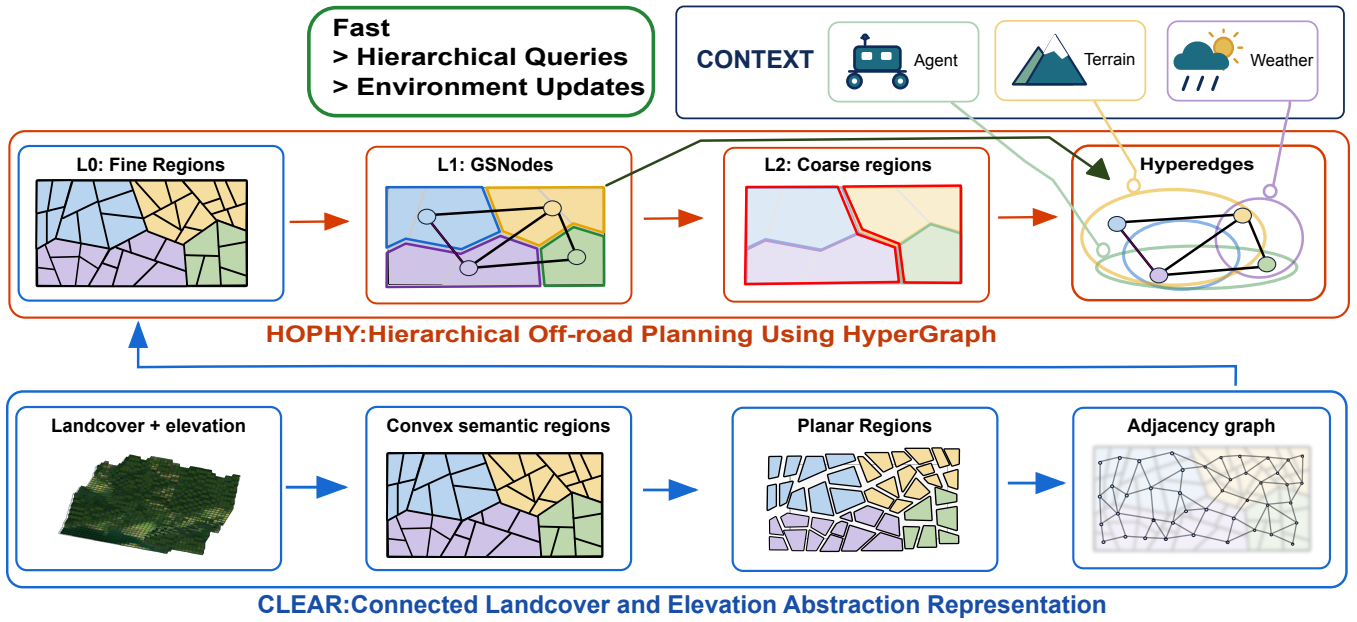


Fig. 1. CLEAR provides the foundational flat abstract representation: convex semantic-geometric regions and their adjacency graph directly support path planning. HOPHY builds a hierarchy over these regions to enable faster path planning and quick localized updates as mission context changes.

select vehicle-dependent feasibility and cost, and mission hyperedges retain operational constraints. Hyperedges act as update operators rather than replacement search spaces. Activating an event changes only affected nodes and incident edges before a new coarse-to-fine query. For a context query q , let $U(q)$ be the matched GSNODE set and $\delta^+(U)$ its incident directed edges. The implementation updates $U(q) \cup \delta^+(U)$ after intersecting the selected membership sets, so work depends on the affected subgraph rather than automatically on the full graph. On the 11,993-GSNODE Wharton graph, varying the affected set from 16 to 4096 nodes yields near-linear empirical exponents of 1.00–1.02 ($R^2 = 0.984$ –1.000) for cost, feasibility, and relabel operations.

III. EVIDENCE ACROSS MAPPING AND PLANNING

Table I reports the evidence most relevant to the combined representation claim. CLEAR establishes scalable construction and reuse of semantic-geometric map units. HOPHY establishes fast hierarchical queries, cost fidelity, and localized event-reactive planning. The evaluations use the same three geographic scales but are conducted independently against the claim of each layer.

HOPHY succeeds on all 270 static path evaluations. In a multi-agent task-allocation loop, its path-cost primitive reduces total computation by $79\times$ relative to pixel A* and $26\times$ relative to the best-performing alternative while retaining comparable mission makespan. Under localized rain and blockage, representation update and replacement planning remain below 200 ms.

A physical Jackal experiment evaluates the real-world feasibility of executing the planned mission. Eight separately recorded routes traverse concrete, road, grass, and sloped muddy terrain for a total of 1505.4 m. Mission generation takes 2.65 s, the median path query takes 108.21 ms, and

TABLE I
EVIDENCE FOR THE TWO REPRESENTATION LAYERS.

Layer	Measure	Result
CLEAR	Query speed vs. raw-grid A*	$2.0\times$ – $19.6\times$
CLEAR	Path-cost overhead vs. A*	6.7%
CLEAR	BeamNG task completion	100%
HOPHY	Median query-time reduction	$68\times$ – $1280\times$
HOPHY	Median cost difference vs. A*	0.00%
HOPHY	Replacement plan after event	< 200 ms
HOPHY	Static planning success	270/270

Nav2 executes the routes in 994.5 s. In the demonstrated M6 blockage scenario, the typed representation update takes 67.5 ms and the replacement query takes 137.1 ms.

IV. CURRENT SCOPE AND FUTURE WORK

The combined representation uses CLEAR’s decomposition to construct the convex fine regions consumed by HOPHY. HOPHY merges connected, semantically compatible fine regions into GSNODEs, organizes those nodes into coarse regions, and associates typed contextual memberships with the resulting planning state. The current scope assumes pre-aligned landcover and elevation, uses dominant labels for mixed regions and manually specified event-to-region grounding, and evaluates physical mission feasibility on one robot platform and one outdoor site.

This boundary motivates a direct extension. An open-vocabulary or multimodal model could ground “wet grass is unsafe for this vehicle” to an explicit region set. A typed hyperedge would activate that set, while a deterministic planner retains responsibility for connectivity and cost. In this way, a semantic label becomes operational when it is tied to geometry, context, and an explicit effect on planning state.

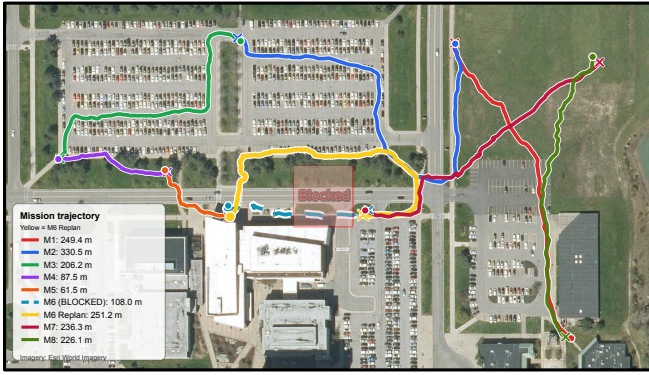


Fig. 2. Jackal mission execution and the M6 blockage case. The replacement route illustrates how maintained semantic terrain state supports contextual replanning during an outdoor mission.

Future work will calibrate uncertainty at region boundaries, learn contextual memberships from multimodal observations, and broaden physical evaluation across environments and robot platforms.

V. CONCLUSION

CLEAR and HOPHY expose two complementary requirements for operational semantic terrain maps. The map units must preserve the landcover and elevation structure that governs mobility, and those units must remain efficiently queryable and locally revisable as mission context changes. CLEAR provides reusable convex semantic–geometric regions. HOPHY provides hierarchical organization and typed contextual updates. Their evaluations show that large outdoor maps can support fast repeated planning, cost-faithful hierarchical search, localized replacement queries, and physical

mission execution. The next step is to connect online multimodal grounding to this explicit region interface so that semantic observations become explicit, localized planning updates rather than unstructured cost changes. This is the central mission-scale takeaway: an operational semantic map persists through the mission as state that can be queried, inspected, and locally adapted.

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